

Dr.M.Usharani

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SUMMARY:

- 27 years of experience in teaching
- One year worked as Communication Engineering trainee at 230 KVSS , Hosur
- Worked as a Production Engineer in single layer PCB company
- **19 papers published in international journals and 1 Paper published in National Journal**
8 published in scopus , 2 published SCI , 6 in UGC, 1 in annexure I journal(AU)
- 22 papers published in national and International conference
- 3 patents published
- Involved in all department major activities
- Anna university recognised research supervisor

RESEARCH IDS

- Anna university research supervisor number ; **4140140**
- ORCID ID <https://orcid.org/0000-0002-5662-1880>
- Scopus ID <https://www.scopus.com/authid/detail.uri?authorId=57210607508>
- Google Scholar Link <https://scholar.google.com/citations?user=8VV67q4AAAAJ&hl=en>
- Web of Science Researcher ID ADG-4093-2022

MEMBERSHIP IN PROFESSIONAL BODIES

- ISTE MISTE Life Time member Membership ID 53108
- IE (I) MIE M-1539345
- IEANG Membership ID 178400
- ISRD : Membership ID M4150902064

PATENT PUBLISHED

S. No	Title	Details of publication	Year of publication
1	Plant and soil monitoring using cloud based IoT for smart agriculture	Application number: 202141038467 A	03/09/2021
2	A smart application for Night Vision Patrolling Robot using Cloud-based- IoT with CNN Design system for Blood Glucose Level	Application number 202141043523 A	01/10/2021
3	Prediction for Diabetics Based on Nutrition and Insulin	Application number 202141057315 A	17/12/2021

GUEST LECTURE DELIVERED

- Webinar delivered on the title of RF Mixer on 19.07.2022 at RVS college of Engineering and Technology-Coimbatore as a resource person.(FDP ON ANTENNAS AND MICROWAVE ENGINEERING).
- Webinar delivered on the title of Optimisation of power management in devices and circuit technology as a resource person in one week Online faculty development program on Low power VLSI and its industry trends 18.04.2022 to 23.04.2022 at Sanskrithi School Of Engineering, Puttaparthi
- On the title Fabrication Process and Layout Design at Engineer Perumal Manimekalai Polytechnic College Hosur
- Webinar delivered on Transmission line and wave guides June 8.6.2020 to 13.6.2020 at Dhanalakshmi Sreenivasan College of Engineering Chennai
- On the title Simulation of Antenna Array and Protocol modelling using XFDTD and NetSim 'on May 6,2016 at Sri Venkateswara college of Engineering College, Chennai
- On the title Basics of electronic circuits and mid band analysis of small signal amplifier at Loyola Institute of Technology on 31.8.2015–Duration 3 hrs
- Conducted national level conferences and symposium . Acted as panel member for project presentations and paper presentations

PAPER PRESENTATION IN CONFERENCES

- 1.M.Usharani, “Statistical Approach for the Development for Driving Support”, National Conference on ‘RecentAdvancements in Communication, Electronics and Signal Processing, 15 Mar, 2019.
- 2.M.Usharani, “Object Detection Tracking of Plantation Crops Using SVM algorithm”, National Conference on ‘Recent Advancements in Communication, Electronics and Signal Processing, 15 Mar, 2019
- 3.M.Usharani, “Detection of Pest in Agricultural Crops by SVM Algorithm”, National level conference on Emerging trends in Engineering Design & Manufacturing ‘ETEDM’-2019, 22 Mar, 2019.
4. M.Usharani, “An Upgraded PENTA MTJ Based Combination AL and Sequential Circuits”, National Conference on ‘Recent Advancements in Communication, Electronics and Signal Processing, Mar, 2018.
5. M.Usharani, “An Enhanced PENTA MJT Based Combinational and Sequential Circuits”, InternationalConferenceon Emerging Trends in Electrical and Communication Technologies – ICETECT - 17, 30-31 Mar, 2018.
6. Brain Metastases from Cancer Using Region Growth Algorithm”, Second International conference on Recent trends in M.Usharani, J.Bhuvaneshwari, V.Niranjana, Monamoirtle, “Identifying the Primary Site of Origin of MRI Science, Engineering and Management, 23-24 Feb, 2018.
- 7.M.Usharani, C.Sathya, “Deployment of Near Field Communication for an Advanced ATM Machine”, National Conference on Modern Electronics and Signal Processing, 15 Mar
8. M.Usharani, K.Janani, M.Revathi, A.Revathi, “High Reliable and Low Cost Hardware Based Chip Identification Generation Based PUF Using Configurable Ring Oscillator”, National Conference on Frontiers in Communication and Signal processing System, 10-11 Mar, 2014.
9. M.Usharani, R.Shankari, “Optimization of FinFETParameters in Multiplier System”, National Conference on Advances in Information and Communication Technology, 27-28 Feb, 2014.
10. M.Usharani, P.Kalaivani, “Design of Gate All Around Tunnel FET for RF Performance”, InternationalConference on Innovations in Intelligent Instrumentation, Optimization & Signal Processing, 01-02 Mar, 2013
11. M.Usharani, Janarthani, “Design of Modified Adaptive Median Filter with PipelinedArchitecture”, International Conference on Modelling,Optimization and Computing, 10-11 Apr, 2012.
12. M.Usharani, R.Vigneshwari, S.Jenbagasundari, M.Valarmathi, S.Krithika, “Hand Gesture Recognition Based Robotic ARM Control and Home Automation”, National Conference Recent Trends in Communication & Signal Processing, 29 Mar, 2012.
13. M.Usharani,Janarthani, “Design of Modified Adaptive Median Filter”, National Conference on VLSI Design and Image Processing, 16 Mar, 2012.

14. M.Usharani, "Lossless Image Compression and Decompression for Grayscale Image Using Context Based Adaptive Lossless Image CODEC", National Conference on Emerging Computing and Communication Technologies, 10-11 Mar, 2011.
15. M.Usharani, P.VigneshPriya, "Image Compression Using Lossless Adaptive Content Based Image Codec", National Conference on VLSI Design and Image Processing, 17 Mar, 2011.
16. M.Usharani, "Emerging Applications of Game theory – A Theoretical Approach", International Conference on Operations Research Applications in Engineering and Management, 27-29 May, 2009.
17. M.Usharani, "An Iris Recognition System to Enhance Security Based on Wavelet Theory", National Conference on VLSI Design and Image Processing, 26 Feb, 2009
18. M.Usharani, "A Model Based Objective Evaluation of Eye Movement Correction in EEG Recordings", National Conference on Advanced Telecommunication Technologies, 19-20 Mar, 2008.
19. M.Usharani, C.Karthikeyan, "Reconfigurable FPGAs and A Network on Chip", National Conference on Soft Computing Techniques in Process Control and Instrumentation, 15-17 Feb, 2007.
20. M.Usharani, "An iris recognition system to enhance a security based on wavelet theory", National Conference on Advanced Techniques in Instrumentation, Control and Communication Engineering, 24 Jan, 2007.
21. M.Usharani, Sridevi, "Error compensation of CSD with sign extension", National conference on signal processing, VLSI and communication held on 17&18 Feb, 2006.
22. M.Usharani, Sridevi, "An iris recognition system to enhance a security based on wavelet theory", National conference on signal processing, VLSI and communication held on 17-18 Feb, 2006.

WORK SHOP ATTENDED :17

SEMINAR ATTENDED : 11

FDP ATTENDED :25

JOURNAL PUBLICATION

1. A novel Cooperative of Cluster Member Nodes with watch dog for Detecting Attacker Nodes G. Theivanathan*, V. P. Kavitha, V. Magesh, M. Usharani Research and Applications: Emerging Technologies Volume 4 Issue 2 HBRP publication
DOI: <https://doi.org/10.5281/zenodo.7074623> 2022
<http://hbrppublication.com/OJS/index.php/RAET/article/view/2630>

2. An Optimized Deep Learning Model Based PV Fault Classification for Reliable Power Generation M. Usharani, V. P. Kavitha, G. Theivanathan, V. Magesh, B. Sakthivel, R. Surendiran SSRG International Journal of Electrical and Electronics Engineering Volume 9 Issue 9, 23-31, September 2022 ISSN: 2348 – 8379 / <https://doi.org/10.14445/23488379/IJEEE-V9I9P103>
© 2022 Seventh Sense Research Group® PP 23 -31 SCOPUS indexed
<https://www.internationaljournalsrg.org/IJEEE/2022/Volume9-Issue9/IJEEE-V9I9P103.pdf>
<https://www.internationaljournalsrg.org/IJEEE/paper-details?Id=38>

3. An Optimized Deep-Learning-Based Low Power Approximate Multiplier Design M. Usharani^{1,*}, B. Sakthivel², S. Gayathri Priya³, T. Nagalakshmi⁴ and J. Shirisha⁵
Computer Systems Science & Engineering, ISSN: 0267-6192 vol. 44, no. 2
DOI: 10.32604/csse.2023.027744 Tech Science Press impact factor 4.397 citation score 1.9 SCI indexed, Scopus indexed 2022
https://file.techscience.com/ueditor/files/csse/TSP_CSSE-44-2/TSP_CSSE_27744/TSP_CSSE_27744.pdf

4. M. Usharani^{1,*}, B. Sakthivel², K. Jayaram³ and R. Renugadevi “Design of Logically Obfuscated Memory and Arithmetic Logic Unit for Improved Hardware Security” Intelligent Automation of Soft Computing, DOI: 10.32604/iasc.2022.023284 ISSN 1079-8587 Tech Science Press SCI indexed, Scopus indexed 2022
<https://www.techscience.com/iasc/v33n3>
https://www.researchgate.net/publication/331938969_List_of_Science_Citation_Index_Expanded_journals_2022_Update_this_month/citation/download

5. Dr. M. Usharani, R. Hemalatha, A. Prithivi, K. B. Samruthi, P. Logeshwari, Hash trick Based deep neural network implementation using optimized stochastic multiplier, Annals of the Romanian Society for Cell Biology, ISSN: 1583-6258, Vol. 25, Issue 6, 2021, Pages. 2075 – 2083 Impact factor (SCOPUS INDEXED JOURNAL) (UGC approved journal)
<https://www.annalsofrscb.ro/index.php/journal/article/view/5751>
<http://www.newswertia.com/download/sci-journals-impact-factor-list-2019-2020/>

6. Dr. M. Usharani, Lavanya .S, Madhumitta .AK. Sinduja .D” Design of circularly polarized patch antenna using polygonal slot” Annals of the Romanian Society for Cell Biology, ISSN: 1583-6258, Vol. 25, Issue 6, 2021, Pages. 1954 – 1963 Impact factor (SCOPUS INDEXED JOURNAL) (UGC approved journal)
<https://www.annalsofrscb.ro/index.php/journal/article/view/5736>
<http://www.newswertia.com/download/sci-journals-impact-factor-list-2019-2020/>

7. Dr. M. Usharani, Mallineni. Jyoshna, B. P. Kruthika, D. Pavithra, Kondapalli. Kedareeswari, “Smart Sewerage Monitoring System” International Research Journal of Engineering and Technology (IRJET), Volume: 07 Issue: 08 | Aug 2020 www.irjet.net p-ISSN: 2395-0072, e-ISSN: 2395-005 Pages– Impact factor 7.529 (UGC approved journal)

8. M.Usharani R.Ishwarya, J.Janani, N.S.Swetha, V.Shanthini, “Hands-Free Operations On Home Appliances Using BCI, International Research Journal of Engineering and Technology (IRJET), Volume: 07 Issue: 08 | Aug 2020 www.irjet.net p-ISSN: 2395-0072, e-ISSN: 2395-0056- Impact factor 7.529 <https://www.irjet.net/archives/V7/i8/Velammal/NCRACES-34.pdf>
9. M.Usharani, S.Ramya, N.Shwetha, Y.Soundarya, VarshaRajkumar, “Object Detection And Tracking Of Plantation Crops Using SVM Algorithm”, International Journal of Applied Engineering Research ISSN 0973-4562 Volume 14, Number 6, 2019 (Special Issue) Impact factor 0.51 (UGC approved journal) https://www.ripublication.com/ijaerspl2019/ijaerv14n6spl_54.pdf
https://www.academia.edu/40012796/List_of_UGC_approved_journals
10. M.Usharani, R.Arulmozhi, A.Devadharshini, “Statistical Approach for the Development for Driving Support System based on Pixel Classification algorithm”, International Journal of Applied Engineering Research, ISSN 0973-4562 Volume 14 Number 6, 2019. Impact factor-0.51 (UGC approved journal) https://www.ripublication.com/ijaerspl2019/ijaerv14n6spl_55.pdf
https://www.academia.edu/40012796/List_of_UGC_approved_journals
11. M.Usharani, J.Bhuvaneswari, V.Niranjana “Recognition of Brain Metastases and Detection of Origin of Cancer using Region Growth Algorithm”, International Journal for Research in Applied Science & Engineering Technology (IJRASET), ISSN: 2321-9653; IC Value: 45.98; SJ Impact Factor: 6.887, Volume 6, Issue III, March 2018. Scientific Journal Impact Factor: 6.887 (UGC approved journal) <https://www.ijraset.com/fileserve.php?FID=14734>
https://www.academia.edu/40012796/List_of_UGC_approved_journals
12. M.Usharani, P.Narmathashri “Optimized Design of Modified Double Tail Comparator”, International Journal of Science Technology and Engineering, ISSN (online): 2349-784X, Volume 4, Issue 2 August 2017. impact factor 4.753 SCOPUS INDEXED JOURNAL <https://www.ijste.org/articles/IJSTEV4I2011.pdf>
13. M.Usharani, Dr.A.Senthilkumar, “Power Optimization of Johnson Counter with and without Split Charge Recovery Logic”, International Journal of Printing, Packaging & Allied Sciences, Vol. 4, No. 5, Dec 2016. Annexur I journal
14. M.Usharani, “High Performance and an Efficient Compensation Methodology in Three stage CMOS operational Transconductance Amplifier (OTA)”, International journal of Innovations in Engineering & Management, ISSN: 2319-3344 volume 5, NO 1, 24 June, 2016. Global impact factor 0.546
15. M.Usharani, C.Sathya “Deployment of Near field Communication for an advanced ATM machine”, International Journal of Applied Engineering Research ISSN 0973-4562 volume 10 No.22(2015), April 2015. Pages – 20504 20507 Impact factor-0.181 (SCOPUS INDEXED JOURNAL) <https://www.ripublication.com/Volume/ijaerv10n5.htm>
https://www.academia.edu/40012796/List_of_UGC_approved_journals
16. M.Usharani, Dr.A.Senthilkumar, “Design and optimization of High Speed Low Power Modified SCRL using QCA circuits”, International Journal of Applied Engineering Research ISSN 0973-4562 volume 10 No.5 (2015) pp. 12855 – 12870, April 2015. Impact factor-0.181 (SCOPUS INDEXED JOURNAL) Annexure II <https://www.ripublication.com/Volume/ijaerv10n5.htm>

https://www.academia.edu/40012796/List_of_UGC_approved_journals

17. M.Usharani, Dr. A. Senthilkumar, R. Shankari, P. Kalaivani “Analysis of Quantum Devices with Conventional Devices”, International Journal of Applied Engineering Research ISSN 0973-4562 volume 10 No.6(2015) pp. 14877 – 14890, April 2015. Impact factor-0.181(SCOPUS INDEXED JOURNAL) Annexure II

<https://www.ripublication.com/Volume/ijaerv10n6.htm>

https://www.academia.edu/40012796/List_of_UGC_approved_journals

18. M.Usharani, C. Sathya “A Survey of Technologies to enable security in near field communication tag design”, International journal of Innovative Research in Advanced Engineering, ISSN 29-2163 volume 1 issue 10 November 2014. Impact factor 1.035

<https://www.ijirae.com/volumes/vol1/issue10/64.NVEC10107.pdf>

19. M.Usharani, P. Kalaivani “Optimization of geometrical parameters of Gate-all-around Tunnel FET for Analog RF Applications”, International Journal of Inventive Engineering and Sciences volume 1 issue 7 June 2013, June 2013

<https://www.ijies.org/portfolio-item/g0236061713/>

<https://www.mendeley.com/search/?page=1&query=Optimization%20of%20%20geometrical%20parameters%20of%20%20Gate-all-around%20Tunnel%20FET%20for%20Analog%20RF%20Applications&sortBy=relevance>

20. M.Usharani, Anitha Julian “An Iris recognition system to enhance a security based on wavelet theory”, Nice Journal of Emerging Technologies volume 3 NO.2 December 2008 pp. 48-57, Dec 2008.